



Islamic University of Technology

Department of Computer Science and Engineering (CSE)

CSE 4206: Digital Logic Design Lab
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Lab 3

Binary Adder Operation

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1 Introduction

In previous labs, we explored the design and implementation of combinational circuits, focusing on fundamental logic gates and their combinations to perform specific tasks. Building upon that foundation, this lab introduces the concept of binary addition using combinational logic. Specifically, we will design and implement binary adders—basic circuits that perform arithmetic addition of binary numbers. The detailed structure, working, and variations such as half adder and full adder will be covered in the experiment.

2 Binary Adder

Adders are important parts of digital electronics and computer systems that help perform addition. Adding binary numbers works just like adding decimal numbers, starting from the rightmost bit (least significant bit). When adding two bits, if the result is more than 1, a **carry bit** is created and passed to the next bit on the left to be added there.

There are two types of adders commonly used in digital circuits:

1. **Half Adder:**

A half adder adds **two single bits** and gives a **sum** and a **carry**.

2. **Full Adder:**

A full adder, on the other hand, adds **three bits**, **two input bits** and an **additional carry** from a previous stage and produces a **sum** and a **carry** output.

3 Overview of Logic Gates and ICs

Gate Name	Logical Expression	IC Number
AND	$A \cdot B$	7408
OR	$A + B$	7432
NOT	$\overline{A}$	7404
NAND	$\overline{A \cdot B}$	7400
NOR	$\overline{A + B}$	7402
XOR	$A \oplus B$	7486
XNOR	$\overline{A \oplus B}$	74266
Full Adder	$\overline{A \oplus B}$	74283

Table 1: Logic Gates and their corresponding ICs (TTL – 74 series)

4 IC Internal Diagram

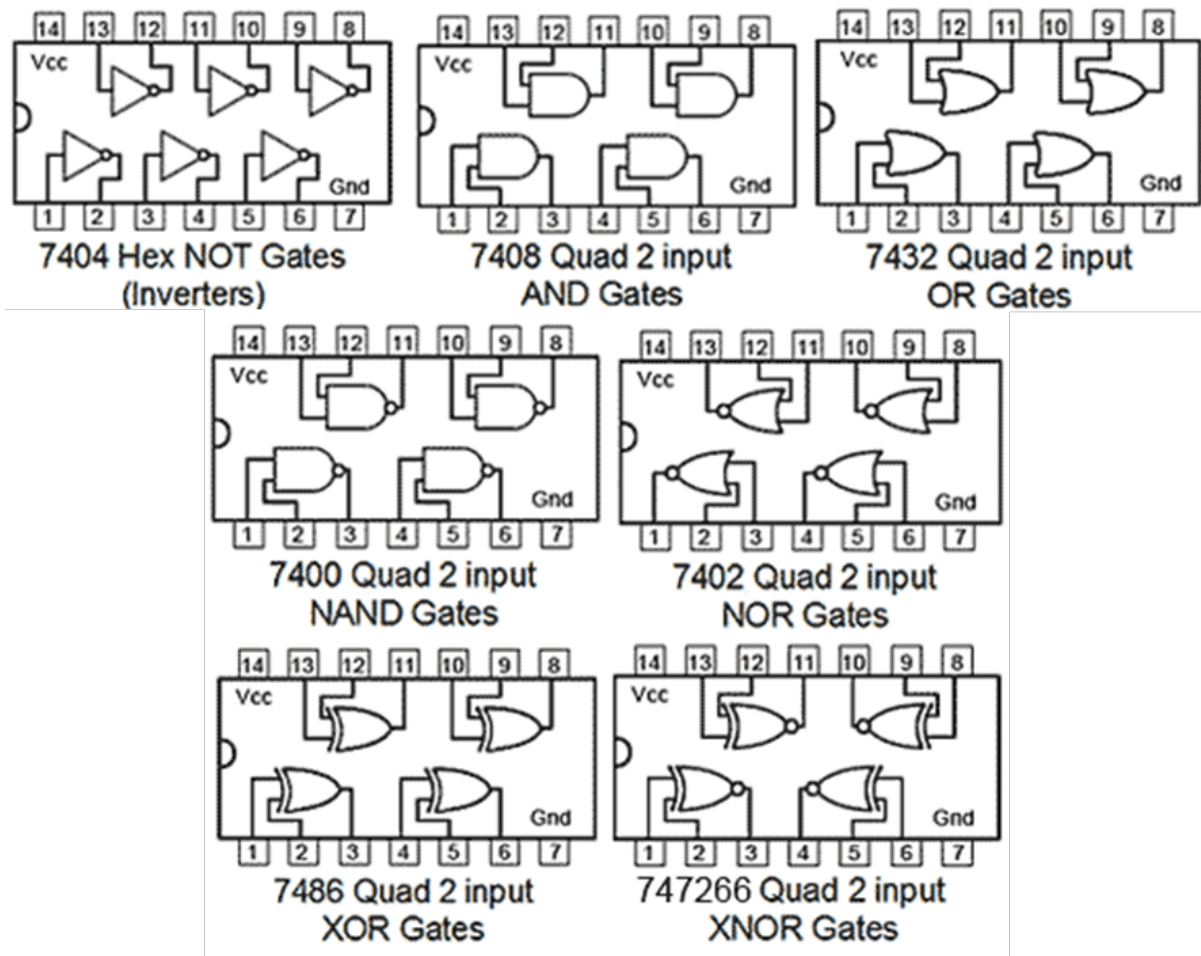


Figure 1: Internal Structure of ICs

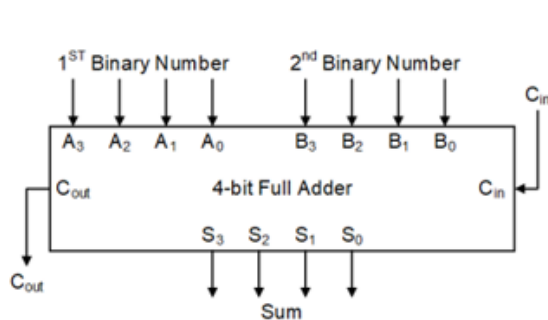


Fig. 1. Logic Symbol for a 4-bit Full Adder

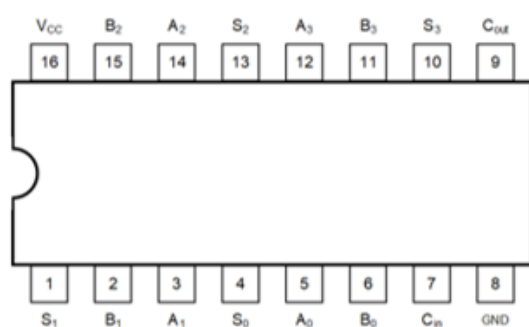


Fig 2. IC 74LS283 Pinout with matching logic labels

Figure 2: Full Adder

5 Experiment 3: Implementation of Adder Circuits

5.1 Objectives

The objectives of this experiment are as follows:

1. To understand and implement binary Half Adder and Full Adder circuits.
2. To design and implement a 4-bit binary adder.

5.2 Tasks

Today's lab task are as follows -

5.2.1 Task 1: Implement 1-bit Half Adder**Tasks**

1. Complete the truth table for a 1-bit Half Adder.
2. Derive the simplified Sum of Products (SoP) expressions for Sum (S) and Carry (C).
3. Draw the Boolean logic circuits for Sum (S) and Carry (C).

Truth Table:

A	B	Sum	Carry
0	0		
0	1		
1	0		
1	1		

Circuit Diagram:

Draw the circuit diagram here using basic gates

Simplified Expression:

Sum, $S =$

Carry, $C =$

Express S using X-OR, $S =$

5.2.2 Task 2: Implement 1-bit Full Adder**Tasks**

1. Complete the truth table for the 1-bit Full Adder.
2. Derive the Sum of Products (SoP) expressions for Sum (S) and Carry (C).
3. Draw the Boolean logic circuits for Sum (S) and Carry (C).

Truth Table:

A	B	C_{in}	Sum	Carry
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

Circuit Diagram:

Draw the circuit diagram here using basic gates

Simplified Expression:

Sum, S =

Carry, C =

Express S using X-OR, S =

Express C using X-OR, C =

5.2.3 Task 3: Implement 2-bit Full Adder

Tasks

1. Implement a 2-bit binary full adder using basic, universal, or exclusive gates.
2. Mathematically show and verify the circuit by performing the binary addition:

$$(3)_{10} + (2)_{10} = (5)_{10}$$

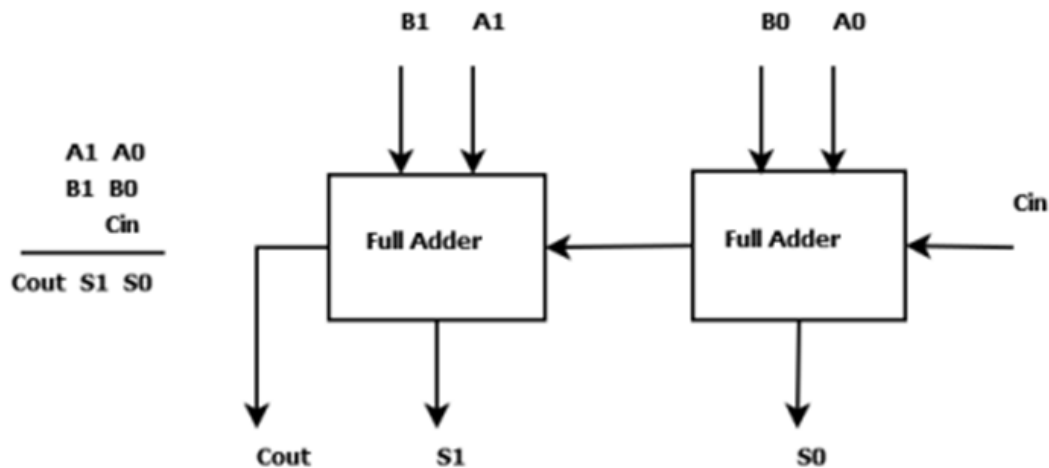


Figure 3: 2 bit Full Adder diagram

5.2.4 Task 4

Tasks Now implement the following tasks using the **4-bit binary full adder IC (74283)**:

1. Implement and verify the task from Lab 3 using the 4-bit binary adder IC.
2. Draw the block diagram for the 4-bit binary full adder.
3. Specify the values of all necessary input and output variables for the operation:

$$(13)_{10} + (9)_{10} = (22)_{10}$$

Provide the values of:

$$A_i, \quad B_i, \quad C_{in}, \quad S_i, \quad C_i$$

where:

- A_i : Bits of input A (from least significant bit A_0 to most significant bit A_3)
- B_i : Bits of input B (from B_0 to B_3)
- C_{in} : Carry input (initial carry, usually 0)
- S_i : Sum output bits (from S_0 to S_3)
- C_i : Carry outputs after each bit addition (C_1, C_2, C_3 , and final carry-out C_4)

4 bit Binary Full Adder Block Diagram:

